COURSE-BASED PROJECT DOCUMENTATION | DOC-14

Build Automation & Execution

Multi-Platform Build Scripts, Compilation, & CLI/GUI Commands

1. Build Automation Overview

The repository provides three automated build pipelines supporting different development environments:

- Windows Command Prompt: build.bat
- Windows PowerShell: build.ps1
- Cross-Platform GNU Make: Makefile

2. Sequential Build Steps

- Step 1: Checks for flex, bison, and gcc in system PATH (injects MSYS2 UCRT64 paths).
- Step 2: bison -v -d parser/java_parser.y --report-file=parser/java_parser.output -o parser/java_parser.tab.c
- Step 3: flex -o lexer/lex.yy.c lexer/java_lexer.l
- Step 4: gcc -O2 -shared -Iparser -Ilexer parser/java_parser.tab.c lexer/lex.yy.c -o javacheck.dll
- Step 5: gcc -O2 -Iparser -Ilexer parser/java_parser.tab.c lexer/lex.yy.c -o javacheck.exe

3. How to Run the System

Launch Graphical User Interface:

```
python gui/app.py
```

Execute Command-Line Syntax Check:

```
# Check valid file
javacheck.exe tests/valid_1.java
# Output: SUCCESS: Valid Java Syntax

# Check invalid file
javacheck.exe tests/invalid_2.java
# Output: ERROR: Line 2: Syntax error near ';'
```

Run Automated Regression Test Suites:

```
python tests/run_tests.py         # Runs 59 parser tests
python tests/test_gui_backend.py  # Runs 14 GUI integration tests
```